

Task#1

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[5];
```

```
    int i, temp;
```

```
    printf("Enter 5 integers: ");
```

```
    for (i = 0; i < 5; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    temp = arr[4]; // Last element
```

```
    for (i = 4; i > 0; i--) {
```

```
        arr[i] = arr[i - 1];
```

```
    }
```

```
    arr[0] = temp;
```

```
    printf("Output: ");
```

```
    for (i = 0; i < 5; i++) {
```

```
        printf("%d ", arr[i]);
```

```
    }
```

```
    printf("\n");
```

```
    return 0;
```

```
}
```

Task#2

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[10];
```

```
    int i, target, count = 0;
```

```
    printf("Enter 10 numbers: ");
```

```
    for (i = 0; i < 10; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    printf("Enter number to search: ");
```

```
    scanf("%d", &target);
```

```
    for (i = 0; i < 10; i++) {
```

```
        if (arr[i] == target
```

```
) {
```

```
        count++;
```

```
    }
```

```
}
```

```
if (count > 0) {
```

```
    printf("Number %d occurred %d times.\n", target, count);
```

```
} else {
```

```
    printf("Number not found.\n");
```

```
}
```

```
return 0;}
```

Task#3

```
#include <stdio.h>
```

```
int main() {
```

```
    int passMarks[10], failMarks[10];
```

```
    int mark, passCount = 0, failCount = 0, i;
```

```
    float passAvg = 0, failAvg = 0;
```

```
    printf("Enter marks for 10 students (enter -1 to exit early):\n");
```

```
    for (i = 0; i < 10; i++) {
```

```
        scanf("%d", &mark);
```

```
        if (mark == -1) break;
```

```
        if (mark >= 5 && mark <= 10) {
```

```
            passMarks[passCount] = mark;
```

```
            passAvg += mark;
```

```
            passCount++;
```

```
        } else if (mark >= 0 && mark <= 5) {
```

```
            failMarks[failCount] = mark;
```

```
            failAvg += mark;
```

```
            failCount++;
```

```
        }
```

```
    }
```

```
    // Display pass marks and average
```

```
    printf("Pass Marks: ");
```

```
    for (i = 0; i < passCount; i++) {
```

```
        printf("%d ", passMarks[i]);
```

```
}  
  
if (passCount > 0) {  
    passAvg /= passCount;  
    printf("\nPass Average: %.2f", passAvg);  
} else {  
    printf("None");  
}  
  
// Display fail marks and average  
printf("\nFail Marks: ");  
for (i = 0; i < failCount; i++) {  
    printf("%d ", failMarks[i]);  
}  
  
if (failCount > 0) {  
    failAvg /= failCount;  
    printf("\nFail Average: %.2f", failAvg);  
} else {  
    printf("None");  
}  
  
printf("\n");  
  
return 0;  
}
```

Task#4

```
#include <stdio.h>
```

```
int main() {
```

```
    char str[100];
```

```
    int i = 0;
```

```
    // Read sentence
```

```
    printf("Enter a sentence: ");
```

```
    scanf("%[^\n]", str);
```

```
    // Convert case, i use ASCII values number of conversion
```

```
    while (str[i] != '\0') {
```

```
        if (str[i] >= 'a' && str[i] <= 'z') {
```

```
            str[i] = str[i] - 32; // To uppercase
```

```
        } else if (str[i] >= 'A' && str[i] <= 'Z') {
```

```
            str[i] = str[i] + 32; // To lowercase
```

```
        }
```

```
        i++;
```

```
    }
```

```
    printf("Converted: %s\n", str);
```

```
    return 0;
```

```
}
```

Task#5

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[10];
```

```
    int i, max, min, diff;
```

```
    printf("Enter 10 integers: ");
```

```
    for (i = 0; i < 10; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    // Find max and min
```

```
    max = arr[0];
```

```
    min = arr[0];
```

```
    for (i = 1; i < 10; i++) {
```

```
        if (arr[i] > max) max = arr[i];
```

```
        if (arr[i] < min) min = arr[i];
```

```
    }
```

```
    diff = max - min;
```

```
    printf("Difference (max - min): %d\n", diff);
```

```
    return 0;
```

```
}
```

Task#6

```
#include <stdio.h>
```

```
int main() {
```

```
    char str[100];
```

```
    int i = 0, vowels = 0, cons = 0;
```

```
    // Read word
```

```
    printf("Enter a word: ");
```

```
    scanf("%s", str);
```

```
    // Count vowels and consonants
```

```
    while (str[i] != '\0') {
```

```
        if ((str[i] >= 'a' && str[i] <= 'z') || (str[i] >= 'A' && str[i] <= 'Z')) {
```

```
            if (str[i] == 'a' || str[i] == 'e' || str[i] == 'i' || str[i] == 'o' || str[i] == 'u' ||
```

```
                str[i] == 'A' || str[i] == 'E' || str[i] == 'I' || str[i] == 'O' || str[i] == 'U') {
```

```
                vowels++;
```

```
            } else {
```

```
                cons++;
```

```
            }
```

```
        }
```

```
        i++;
```

```
    }
```

```
    // Display total vowels and cons
```

```
    printf("Vowels: %d\n", vowels);
```

```
    printf("Consonants: %d\n", cons);
```

```
    return 0;
```

```
}
```

Task#7

```
#include <stdio.h>

int main() {

    int arr[10];

    int i, j;

    // take 10 integers
    printf("Enter 10 integers: ");

    for (i = 0; i < 10; i++) {

        scanf("%d", &arr[i]);

    }

    // replace duplicates with -1
    for (i = 0; i < 10; i++) {

        if (arr[i] != -1) {

            for (j = i + 1; j < 10; j++) {

                if (arr[j] == arr[i]) {

                    arr[j] = -1;

                }

            }

        }

    }

    printf("Updated array: ");

    for (i = 0; i < 10; i++) {

        printf("%d ", arr[i]);

    }

    printf("\n");

    return 0;

}
```

Task#8

```
#include <stdio.h>
```

```
int main() {
```

```
    char str[100];
```

```
    int i = 0;
```

```
    printf("Enter non-alphabetic characters: ");
```

```
    scanf("%[^A-Za-z]", str);
```

```
    printf("Non-alphabetic characters: %s\n", str);
```

```
    return 0;
```

```
}
```